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Metallic properties of $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ (Pn=As, Sb)

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ABSTRACT

We report the synthesis of ternary barium copper pnictides, $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ (Pn=As, Sb), and their structural, magnetic, and transport properties. They all crystallize in different structures shown by X-ray diffraction, although their structures reveal close relations. The body-centered tetragonal $BaCu_2As_2$ adopts $ThCr_2Si_2$-type ($I4/mmm$) structure, whereas $Ba_2Cu_3P_4$ is a copper-deficient derivative of this phase, crystallizing in the body-centered orthorhombic space group, *Ibam*. There are two polymorphs of $BaCu_2Sb_2$: α-$BaCu_2Sb_2$ that adopts $CaBe_2Ge_2$-type structure; β-$BaCu_2Sb_2$ that is a 2:1 combination of $CaBe_2Ge_2$- and $ThCr_2Si_2$-type structural segments. All phases are metallic and non-magnetic. The room temperature thermal conductivity for polycrystalline $BaCu_2As_2$ is ≈ 2 W/(m K) and the Seebeck coefficient is ≈ 15 μV/K, which result in a small (≈0.03) thermoelectric figure of merit.

1. Introduction

The discovery of superconductivity in the ternary pnictides and chalcogenides with tetragonal structures [1–3] fueled international interest for finding superconductivity in similar iron-based compounds with LaFePnO, LiFePn, and $BaFe_2Pn_2$ with pnictogen elements (Pn). Such work has mainly concentrated on arsenides, and to a lesser extent, phosphides. The antimonide analogs such as LaFeSbO or $BaFe_2Sb_2$ are not prepared so far but there is a great incentive. This is because LaFeAsO is found to have a higher critical superconducting temperatures (T_c) compared to the smaller pnictide analog, LaFePO, and superconductivity in LiFeSb is theoretically predicted [4] at higher temperatures compared with LiFeAs (T_c=18 K) [5]. The research on iron-based superconductors was subsequently extended into systems containing other 3d transition metals. The chemical doping experiments on the antiferromagnetic $BaFe_2As_2$ led to superconductivity by electron doping using cobalt (T_c=22 K) [6] but not hole doping using chromium [7]. The chemical substitution of Fe with other 3d transition metals resulted in a variety of behaviors including itinerant antiferromagnetism in $BaCr_2As_2$ [8], antiferromagnetic insulator in $BaMn_2As_2$ [9], correlated metal in $BaCo_2As_2$, and superconductivity in $BaNi_2As_2$ ($T_c \approx 0.6$ K) [10]. Theoretically, $BaCu_2As_2$ is metallic and nonmagnetic, with low-lying Cu d states [11]. The choice of transition metal not only influences the physical properties, but also chemical bonding in these phases. The transition metals with more electrons in the 3d orbitals give rise to formation of homoatomic Pn–Pn bonds that connect two adjacent layers, and ultimately, turn the structure from 2D into the 3D network [12]. The formation of Pn_2 dimers is an important factor, which influences the crystal and electronic structures, magnetism, and presence of superconductivity in the ternary pnictides with $ThCr_2Si_2$-type structures [13–16].

Our study here focuses on the barium copper pnictide systems $BaCu_2Pn_2$, with Pn=As and Sb. According to the earlier studies, $Ba_2Cu_3P_4$ [17] is one forth Cu deficient derivative of the $ThCr_2Si_2$-structure, $BaCu_2As_2$ crystallizes with $ThCr_2Si_2$ structure, while $BaCu_2Sb_2$ [18] is composed of $ThCr_2Si_2$- and $CaBe_2Ge_2$-type fragments in 1:2 ratio (Figs. 1 and 2). Although the physical properties of these phosphide and arsenide compounds are not known, β-$BaCu_2Sb_2$ is a metal with temperature-independent diamagnetic behavior [19]. We report synthesis, electrical resistivities, magnetization, single crystal and powder X-ray diffraction for $BaCu_2As_2$, $BaCu_2Sb_2$, and $Ba_2Cu_3P_4$. We also find the thermal transport properties and Seebeck coefficients for $BaCu_2As_2$. Our exploratory synthesis in Ba–Cu–Sb system leads to a new polymorph of $BaCu_2Sb_2$, we call α modification, which crystallizes in $CaBe_2Ge_2$-type structure. We also find evidence for existence of another polymorph of $BaCu_2Sb_2$ with $ThCr_2Si_2$-type structure, and possibly other metastable polymorphs containing various combinations of $CaBe_2Ge_2$- and $ThCr_2Si_2$-type layers.

2. Experimental

The elemental reactants (99.9% purity or higher), dendritic Ba pieces, Cu powder, P lumps, As lumps, and Sb shots were all from Alfa Aesar. Prior to its use, copper was reduced in a diluted

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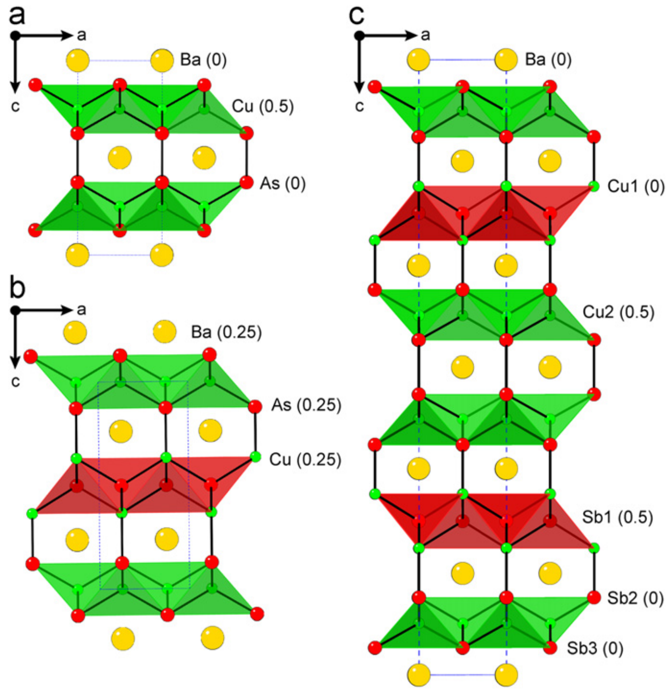

Fig. 1. Polyhedral representations of the crystal structures of tetragonal (a) $BaCu_2As_2$, (b) α-$BaCu_2Sb_2$, and (c) β-$BaCu_2Sb_2$, all viewed down the *b*-axis. The polyanionic $[Cu_2Pn_2]^{2-}$ layers are emphasized; unit cells are outlined. Color on the Web: Ba, Cu and Pn atoms are drawn as yellow, green and red spheres, respectively. Out of plane (*y*) coordinates are shown in brackets next to atom names. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

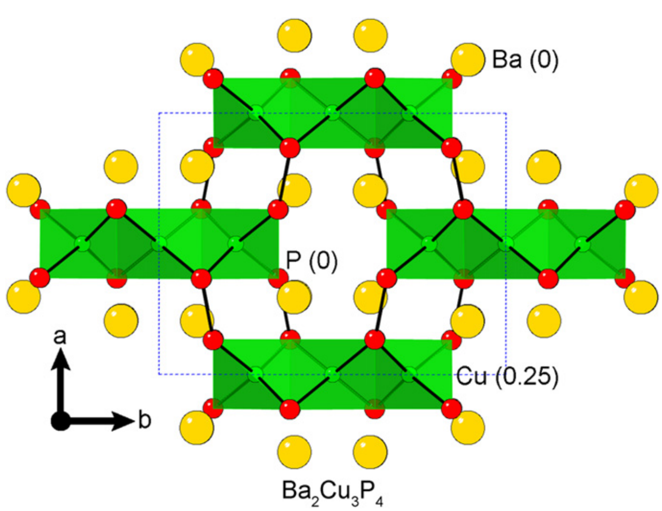

Fig. 2. A representation of the body-centered orthorhombic structure of $Ba_2Cu_3P_4$ viewed down the *c*-axis. Color on the Web: Ba, Cu and P atoms are drawn as yellow, green and red spheres, respectively. In brackets next to atom names, out of plane (*z*) coordinates are shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

H_2 stream (4/96% H_2/Ar) overnight, whereas other elements were used as received. Single crystals of $Ba_2Cu_3P_4$ were produced from a melt flux reaction of Ba:Cu:P:Sn = 2:3:4:15. Single crystals of $BaCu_2As_2$ and $BaCu_2Sb_2$ were each produced using a melt flux reaction of Ba:Cu:Pn:Pb = 1:2:2:10. These compositions were placed in alumina crucibles, and sealed inside silica tubes under a 2/3 atm of ultra-high purity argon to suppress the vapor pressures of arsenic and phosphorus (reactions aimed at antimonides were vacuum sealed). All reaction mixtures were slowly (50 °C/h) heated to 250 °C, then heated to 950 °C at a rate of 200 °C/h, kept at this temperature for 20–24 h, then cooled to 550 °C at a rate of 5–10 °C/h. At this temperature, the flux was mostly separated from products by centrifugation. The products were needle-shaped crystals of $Ba_2Cu_3P_4$, and plate crystals of $BaCu_2As_2$ and α-$BaCu_2Sb_2$, a new polymorph of $BaCu_2Sb_2$. In order to explore polymorphism in $BaCu_2As_2$ and $BaCu_2Sb_2$, the compounds were also synthesized from stoichiometric melts, which were placed in alumina crucibles and sealed inside silica tubes. The reactions were heated to 600 °C at a rate of 30 °C/h, annealed at this temperature for 10 h, then heated to 1050 °C at a rate of 60 °C/h, kept at this temperature for 5 h, finally cooled to 550 °C at a rate of 4.5 °C/h. The reactions were subsequently cooled to room temperature by turning off the furnace. The products could easily be cleaved into plates. In case of the $BaCu_2Sb_2$ reaction, the product contained a mixture of multiple phases including β-$BaCu_2Sb_2$ and other polymorphs.

Room temperature X-ray diffraction patterns were collected on a PANalytical X'Pert PRO MPD X-ray Powder Diffractometer using the Ni-filtered Cu-K_α radiation. Due to a strong overlap of some of the peaks, data collections were repeated using an incident beam Johansson monochromator. A typical run aimed at phase identification was performed in 5–65° (2θ) range, with a step size of 1/60° and ~20 s/step counting time. All samples were confirmed to be stable under ambient air for months, as no change in powder X-ray patterns were observed in this time frame. Longer runs aimed at refinements of the crystal structures and chemical compositions were carried out in 15–65° (2θ) range, with a step size of 1/60° and a counting time of 300–600 s/step in a continuous scan type. GSAS [20,21] software package was used for the Rietveld refinement of the unit cell parameters and atomic coordinates of $BaCu_2As_2$ and α-$BaCu_2Sb_2$. Refinement of structural parameters of melt grown $BaCu_2Sb_2$ was not possible due to the presence of multiple modifications of this phase (β-from, γ-form etc.), with many overlapping peaks.

Selected under a microscope, single crystals of $BaCu_2As_2$ and α-$BaCu_2Sb_2$ were cut into suitable dimensions (all sides less than 0.1 mm), and transferred onto a Bruker SMART APEX CCD-based diffractometer, which uses fine-focus Mo K_α radiation (λ = 0.71073 Å). The crystals were covered in Paratone N oil and kept under a stream of liquid nitrogen at 173(2) K. The structures were solved by direct methods and refined by full matrix least-squares methods on F^2 using the SHELXTL software package [22]. Semi-empirical absorption correction was applied using SADABS within the SHELXTL package. Occupancy for each position was checked by freeing individual site occupation factors (SOF). In the last step, all sites were refined anisotropically and the atomic coordinates were standardized following those in literature. Important data collection and structure refinement parameters are provided in Table 1; atomic coordinates and equivalent isotropic displacement parameters and selected interatomic distances are listed in Tables 2 and 3, respectively, for $BaCu_2As_2$ and α-$BaCu_2Sb_2$. Further details of the crystal structure studies are summarized in the form of CIF (Crystallographic Information File) files [23].

Scanning electron microscope and energy-dispersive X-ray spectroscopy (EDS) measurements on the crystals were carried out using a Hitachi-TM3000 microscope with a Bruker Quantax 70 EDS system. The $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ compositions were confirmed in each of the corresponding reaction mixtures.

Four-probe electrical resistivity, Seebeck coefficient and thermal conductivity measurements were performed on a Quantum Design Physical Property Measurement System (PPMS). Electrical and thermal connections to the selected crystals were made using platinum wires and Dupont 4929N silver paste or Epotek H20E silver epoxy. Typical measurements were performed down to 2 K and up to 400 K. Within the method used for thermal conductivity

Table 1
Selected single crystal data and structure refinement parameters for $BaCu_2As_2$ and α-$BaCu_2Sb_2$.

Empirical formula	$BaCu_2As_2$	α-$BaCu_2Sb_2$
Formula weight (g/mol)	414.26	507.92
Collection temperature (K)	173(2)	
Radiation, wavelength (Å)	Mo K_α, 0.71073	
Crystal system	Tetragonal	
Space group, Z	$I4/mmm$, 2	$P4/nmm$, 2
a, Å	4.4119(4)	4.6536(3)
c, Å	10.193(2)	10.7946(13)
Cell volume, (Å^3)	198.41(4)	233.77(4)
Density (calculated, g/cm^3)	6.934	7.216
Absorption coefficient (mm^{-1})	36.742	28.447
Reflections collected	1115	3069
Independent reflections	94	211
R^a indices (all data)	$R_1=0.0235$	$R_1=0.0231$
	$wR_2=0.0584$	$wR_2=0.0646$
Goodness-of-fit on F^2	1.107	1.300
Largest diff. peak/hole (e$^-$/Å^3)	1.17/−1.65	1.08/-2.01

[a] $R_1=\sum||F_o|-|F_c||/\sum|F_o|$; $wR_2=|\sum|w(F_o^2-F_c^2)^2||^{1/2}$, where $w=1/|\sigma^2F_o^2+(AP)^2+BP|$, and $P=(F_o^2+2F_c^2)/3$; A and B are weight coefficients.

Table 2
Atomic coordinates and equivalent isotropic (U_{eq} [a]) displacement parameters for $BaCu_2As_2$ and α-$BaCu_2Sb_2$.

Atom	Wyckoff site	x	y	z	U_{eq} (Å^2)
$BaCu_2As_2$					
Ba	2a	0	0	0	0.0095(4)
Cu	4d	0	0.5	0.25	0.0115(5)
As	4e	0	0	0.37395(8)	0.0117(4)
α-$BaCu_2Sb_2$					
Ba	2c	0.25	0.25	0.23876(7)	0.0106(3)
Cu1	2a	0.75	0.25	0	0.0145(4)
Cu2	2c	0.25	0.25	0.63189(15)	0.0137(4)
Sb1	2b	0.75	0.25	0.5	0.0083(3)
Sb2	2c	0.25	0.25	0.87380(8)	0.0110(3)

[a] U_{eq} is defined as one third of the trace of the orthogonalized U_{ij} tensor.

Table 3
Selected interatomic distances (Å) in $BaCu_2As_2$ and α-$BaCu_2Sb_2$.

$BaCu_2As_2$			α-$BaCu_2Sb_2$		
Atom pair		Distance	Atom pair		Distance
Ba–	Cu × 8	3.3705(4)	Ba–	Cu1 × 4	3.4722(6)
	As × 8	3.3739(4)		Cu2 × 4	3.5746(7)
				Sb2 × 4	3.5077(4)
				Sb1 × 4	3.6560(6)
Cu–	As × 4	2.5421(4)	Cu1–	Sb2 × 4	2.6963(4)
As–	As	2.570(2)	Cu2–	Sb1 × 4	2.7278(8)
			Cu2–	Sb2	2.611(2)

measurement in PPMS, it is well established that radiative heat losses between the sample and the shoe assemblies, and the surrounding environment result in an overestimation of thermal conductivities at temperatures above 200 K. Some of the heat produced radiates instead of travelling through the sample, which gives a “tail” with a T^3 temperature dependence. Sample preparation for the measurements of Seebeck coefficient and thermal conductivity was as follows: a melt-grown chunk of $BaCu_2As_2$ was ground and pressed into a pellet (cold-pressing) and then annealed at 500 °C for 2 days.

DC magnetization measurements as a function of temperature were carried out below 100 K using a superconducting quantum interference device magnetometer. The applied field of up to 1 T was used.

3. Results and discussion

3.1. Crystal structures

The crystal structures of $BaCu_2As_2$, α-$BaCu_2Sb_2$ and β-$BaCu_2Sb_2$ are shown in Fig. 1. Although the phases are isoelectronic and their molar elemental ratios are identical, they adopt different crystal structures. $BaCu_2As_2$ crystallizes with $ThCr_2Si_2$-type structure in the body-centered tetragonal space group $I4/mmm$ (Pearson symbol $tI10$), whereas α-$BaCu_2Sb_2$ crystallizes with $CaBe_2Ge_2$-type structure in the primitive tetragonal space group $P4/nmm$ (Pearson symbol $tP10$). These two structures are closely related, as it can be inferred from Figs. 1a and b. This fact also can be deduced from the group–subgroup relationship between the space groups $I4/mmm$ (No. 139) and $P4/nmm$ (No. 129), according to which the latter is a *klassengleiche* (k) subgroup of the former. Consequently, the Cu (Wyckoff site 4d) and As (Wyckoff site 4e) positions in $BaCu_2As_2$ are split into two separate crystallographic sites which are separately occupied by Cu and Sb atoms in α-$BaCu_2Sb_2$ structure. This leads to five unique crystallographic sites in the asymmetric unit of α-$BaCu_2Sb_2$ structure, compared to only three unique positions in $BaCu_2As_2$. Alternatively, the $BaCu_2As_2$ structure can be described as 2D layers of $[Cu_2As_2]^{2-}$ made of $CuAs_4$ tetrahedra and barium cations filling the voids in between. In α-$BaCu_2Sb_2$, the layers are alternating — one layer made of Cu-centered $CuSb_4$ tetrahedra, and the next made of Sb-centered $SbCu_4$ tetrahedra.

The new α-$BaCu_2Sb_2$ is reported here for the first time, although it is not a surprising finding. In literature, several antimonides are reported with $CaBe_2Ge_2$ structure (e.g., $EuCu_2Sb_2$, $SrCu_2Sb_2$) as opposed to their arsenide $ThCr_2Si_2$ structure counterparts (e.g., $EuCu_2As_2$, $SrCu_2As_2$) [18,24], presumably caused by electronic and size effects. In fact, in $EuCu_2As_{2-x}Sb_x$ solid solution, a transition from one structure type to another occurs when $x=1.35$ [18]. In $BaCu_2Sb_2$, we observe polymorphism in the parent compound, without any elemental substitution into the structure. Polymorphism has also been reported in compounds such as APd_2Pn_2 (where A=Sr, Ba, and Pn=As, Sb) [25] and

$LaIr_2Si_2$ [26]. Also, superconductivity has been reported for the high temperature modification of $LaIr_2Si_2$ (T_c=1.6 K), which adopts $CaBe_2Ge_2$-type structure, whereas no superconductivity was observed for the low temperature $ThCr_2Si_2$ form [26]. Authors related the physical properties to structural differences, where within the $CaBe_2Ge_2$ structure, half of Ir atoms have a square pyramidal coordination of Si atoms, while in $ThCr_2Si_2$ structure, only tetrahedrally coordinated Ir atoms are present [26].

The structure of the previously known polymorph, β-$BaCu_2Sb_2$, is shown in Fig. 1(c). As described above, the α-modification is product of a flux reaction, whereas the thermodynamically more stable β-form is synthesized from direct reaction of elemental reactants. The fact that the latter is the more stable modification is evidenced by transformation of α- into β-structure upon heating at 500 °C for three days (see Fig. 3). Melt-grown $BaCu_2Sb_2$ samples, however, cannot be indexed by β-form alone — peaks in the diffraction pattern are doubled and tripled, clearly visible in the higher angle region. We were puzzled by the presence of these split peaks, and initially attributed it to a secondary phase, and tried annealing at 500 °C, but even after this heat treatment, the peaks were present. The EDS measurements of crystals from this and all other reactions indicated compounds with 1:2:2 ratios as the only products of the respective reactions, therefore, we concluded that the secondary phase is most likely another polymorph of $BaCu_2Sb_2$. In an attempt to explain this result, we carried out single crystal X-ray diffraction experiments on multiple crystals from the melt-grown samples. However, suitable crystals for single crystal X-ray diffraction could not be found, presumably due to plates not being perfectly aligned on top of each other and/or plates being an intergrowth of multiple layered phases. Through short data collections aimed at unit cell determination, we were able to extract the unit cell parameters reported for the β-form in literature (body-centered tetragonal cell, $a=b\approx 4.63$ Å, and $c\approx 32.60$ Å). For a different crystal from the same batch, another set of lattice parameters (body-centered tetragonal cell, $a=b\approx 4.63$ Å, and $c\approx 10.88$ Å), which has not been reported in literature so far, was found. The latter indicates a phase (we will label this phase as the γ-form) with the $ThCr_2Si_2$ structure. The majority of peaks in the powder diffraction data (Fig. 3(c)) could be assigned to those originating from the β- and γ-forms; however, due to a strong overlap of peaks, an unequivocal statement cannot be made. Considering the fact that β-$BaCu_2Sb_2$ is a 2:1 (*ABA*) combination of Ca Be_2Ge_2-type (*A*) and $ThCr_2Si_2$-type (*B*) structures, we believe that there could be different combinations of layers going from α-$BaCu_2Sb_2$ ($CaBe_2Ge_2$-type) to γ-$BaCu_2Sb_2$ ($ThCr_2Si_2$-type) structures. However, a distortion of the tetragonal $BaCu_2Sb_2$ to monoclinic structure may also explain additional peaks as recently reported for α-$SrRh_2As_2$ [27] and argued for $BaCu_2Sb_2$ [19]. A dedicated work aimed at understanding structural phase relations between various modifications of $BaCu_2Sb_2$ is required for a more concrete and detailed analysis.

No polymorphism was observed for $BaCu_2As_2$ — crystals adopt the $ThCr_2Si_2$ structure irrespective of the synthetic method employed. Furthermore, the compound remains in this structure after multiple heat treatments. No impurity has been detected in the samples synthesized from both the flux (although traces of the Pb flux may remain on top of the crystals) and stoichiometric reactions. The lattice parameters for $BaCu_2As_2$ from Rietveld refinement of the powder X-ray patterns are $a=b=4.42122(2)$ Å and $c=10.22002(4)$ Å within the space group $I4/mmm$ (R_p=1.43%, wR_p=1.93% and χ^2=2.282). Many ternary copper arsenide structures have been reported with partially occupied copper sites [24], however, in our samples, no noticeable deviation from stoichiometry was observed based on both single crystal and powder X-ray diffraction results. The occupancies of copper sites were checked for both $BaCu_2As_2$ and α-$BaCu_2Sb_2$ using the single crystal X-ray data. Although $BaCu_2As_2$ is found to be stoichiometric, α-$BaCu_2Sb_2$ showed a possible slight copper deficiency on Cu1 position. However, the SOF is higher than 0.96, which makes a definitive claim impossible. The refinements of Cu1 occupancy in α-$BaCu_2Sb_2$ did not improve reliability factors (*R*-values) or difference Fourier map (Table 1), and only the equivalent isotropic displacement parameter was slightly lowered to 0.0124(6) Å^2 (Table 2). We note that in β-$BaCu_2Sb_2$ [18], a copper deficiency with a SOF of 0.97 was reported, although the thermal displacement parameters remained elevated for Cu2 atom. Together with the polymorphism in $BaCu_2Sb_2$, the possibility of phase width further complicates understanding of the structural phase transitions in this system. According to EDS measurements, the compositions showed no significant deviation from stoichiometry.

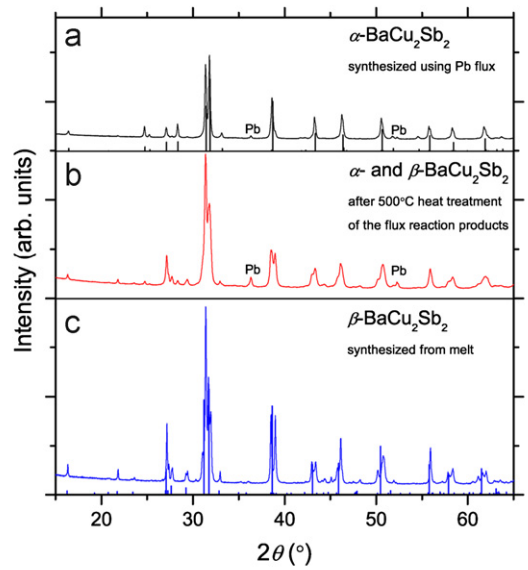

Fig. 3. The powder X-ray diffraction patterns for $BaCu_2Sb_2$: (a) α-phase, (b) a sample containing α- and β-phases and (c) β-phase. Theoretical powder patterns for α-form and β-form are shown as black and blue bars, respectively. See text for more details. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Our repeated attempts to synthesize $BaCu_2P_2$ and $BaCu_2Bi_2$ using various metal fluxes failed. To best of our knowledge, no ternary copper bismuthide with this composition has been reported so far. Although $BaCu_2P_2$ is not reported, $Ba_2Cu_3P_4$ is a known compound [17]. Its crystal structure, shown in Fig. 2, can be rationalized easily if the formula is simplified to $BaCu_{1.5}P_2$ form. Removal of every forth Cu atom from the [Cu_2P_2] layer in $ThCr_2Si_2$-type structure yields [$Cu_{1.5}P_2$] triple chains, which are further interconnected to form a 3D network. This phase was obtained every time we attempted to prepare $BaCu_2P_2$, and in turn, $Ba_2Cu_3As_4$ and $Ba_2Cu_3Sb_4$ could not be synthesized from melt solution despite multiple attempts. According to literature [24], even low partial occupancy of Cu atom in phases with heavier pnictogen atoms or smaller alkaline-earth metals do not lead to this structure type. For example, $EuCu_{1.75}As_2$ [18] and $CaCu_{1.75}P_2$ [28] are known compounds with $ThCr_2Si_2$-type structures.

The results of single crystal X-ray diffraction measurements on $BaCu_2As_2$ and α-$BaCu_2Sb_2$ are summarized in Tables 1–3. The bond distances are in range of the values reported in literature [24]. The interlayer separations are quite short, and their values are given by homoatomic As–As bond distance $d_{(As-As)}=2.570(2)$ Å for $BaCu_2As_2$ and by Cu–Sb bond distance $d_{(Cu2-Sb2)}=2.611(2)$ Å for α-$BaCu_2Sb_2$. Interestingly, in $Ba_2Cu_3P_4$, the inter-chain distances are even smaller $d_{(P-P)}=2.236$ Å, whereas in β-$BaCu_2Sb_2$, there are two types of interlayer distances, $d_{(Cu1-Sb2)}=2.623$ Å and $d_{(Sb3-Sb3)}=2.878$ Å, originating from $CaBe_2Ge_2$-type and $ThCr_2Si_2$-type motifs, respectively. Considering the fact that our data for α-$BaCu_2Sb_2$

were collected at 173(2) K, the agreement in bond distances in α- and β-$BaCu_2Sb_2$ is remarkable, as if there has been no change to the electronic states of the different structural sub-units in the β-modification.

The features of the crystal structures of $BaCu_2As_2$ and $BaCu_2Sb_2$ could be rationalized by the in-depth theoretical work on $ThCr_2Si_2$- [12] and $CaBe_2Ge_2$-type [29] structures. The work on $ThCr_2Si_2$-type structure [12] predicts formation of strong covalent Pn–Pn bonds going from Mn to Cu among the 3d transition metals, and indeed, in our case, such bonds are present. The preference between $ThCr_2Si_2$- and $CaBe_2Ge_2$-type structures has been attributed to a number of factors [29]. One of the outlined trends is that heavier anions within the same group result in $CaBe_2Ge_2$-type structure, for example, going from $BaMg_2Ge_2$ to $BaMg_2Pb_2$, the structure changes from $ThCr_2Si_2$-type to $CaBe_2Ge_2$-type. Similar trend is observed between $BaCu_2As_2$ and $BaCu_2Sb_2$, with the latter crystallizing in $CaBe_2Ge_2$-type structure. In our case, the preferred $ThCr_2Si_2$ structure of $BaCu_2As_2$ could also be understood by a simple reasoning based on the electronegativities of Cu and As: The more electronegative As atoms prefer the outer site, surrounding the more electropositive Cu atoms in a tetrahedral environment. The electronic structure calculations on $CaBe_2Ge_2$ [29] suggest presence of two different layers, which have different band gaps, and consequently, this material exhibits a periodic variation of the band gap in the lattice. This would suggest that in β-$BaCu_2Sb_2$ such band gap variation is even more complex as the periodicity of the occurrence of the two different layers is different, which is implied by the large *c*-lattice parameter.

3.2. Physical properties

Property measurement results are shown in Figs. 4–6. $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ (Pn=As, Sb) are all diamagnetic, giving similar temperature-independent magnetization result to that of α-$BaCu_2Sb_2$ (Fig. 4). In the case of α-$BaCu_2Sb_2$, magnetization measurements were performed both parallel and perpendicular to the *c*-axis. The *c*-axis magnetic susceptibility is roughly an order of magnitude higher than that in the *ab*-plane. For the similar $SrCu_2Sb_2$ compound, the ratio of susceptibilities along the two directions was found to be $\chi_c/\chi_{ab} \approx 4$ [19]. Theory predicted $BaCu_2As_2$ to be far from magnetism [11], which we now confirm experimentally. A simple electron count, with an assumption of ionic interactions between Cu and P, suggests presence of divalent Cu^{2+} (d^9) cations in $Ba_2Cu_3P_4$ according to $(Ba^{2+})_2(Cu^{2+})(Cu^{1+})_2(P^{2-})_4$, but we find that such a simplistic charge assignment is invalid in this case.

The temperature dependences of electrical resistivity for $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ are shown in Fig. 5. The measurements were carried out on single crystals flux grown $Ba_2Cu_3P_4$, $BaCu_2As_2$, α-$BaCu_2Sb_2$ and melt grown $BaCu_2Sb_2$. From the X-ray diffraction experiments, we inferred presence of multiple phases in the melt grown sample of $BaCu_2Sb_2$ (possibly other polymorphs consisting of different combinations of $CaBe_2Ge_2$-type and $ThCr_2Si_2$-type structures, see above). The resistivity (ρ) decreases upon cooling from room temperature values of 0.08–0.15 mΩ cm almost linearly down to less than 0.04 mΩ cm, however in low temperature region (<25 K), there is a slight change in slope due to the residual resistivity from the defect scattering. The numerical values of the room temperature resistivities are 1–2 orders of magnitude higher than that of Cu metal (ρ_{Cu}(298 K)=0.00168 mΩ cm). The obtained results on $BaCu_2As_2$ are in agreement with the electronic structure calculations on $SrCu_2As_2$ and $BaCu_2As_2$, which predicts these compounds to be *sp* metals [11]. In the case of α-$BaCu_2Sb_2$ ($CaBe_2Ge_2$-type) the compound exhibits metallic behavior as well. The electronic structure calculations suggest metallic behavior for $CaBe_2Ge_2$ as well [29], wherein the Fermi level crosses partially (7/8) filled bonding states made of s and p orbitals of Be and Ge. In analogy to the treatment of $CaBe_2Ge_2$ as a sum of two different structural segments [29], β-$BaCu_2Sb_2$ could also be considered as a sum of $ThCr_2Si_2$- and $CaBe_2Ge_2$-type slabs, which are both metallic. Therefore, the combination of these slabs in β-$BaCu_2Sb_2$ may be expected to result in a metallic behavior. Although not a direct proof of this conjecture, our resistivity measurement on the melt-grown $BaCu_2Sb_2$ plates indicate metallic behavior. No theoretical work on $Ba_2Cu_3P_4$ has been conducted so far, we find metallic behavior for this phase as well. There is no apparent trend in the electrical resistivities of the studied phases due to the different crystal and therefore electronic structures of the compounds. The residual resistivity ratios (RRR=ρ(300 K)/ρ(4 K)) are ≈38 for $Ba_2Cu_3P_4$, ≈4 for $BaCu_2As_2$, ≈2.1 for α-$BaCu_2Sb_2$ and ≈22 for the melt grown $BaCu_2Sb_2$, suggesting that the studied crystals are of reasonably good quality. Comparing the results of measurements on single crystals along the *ab*-plane (Fig. 5) and the polycrystalline data (Fig. 6), electrical resistivity anisotropy is roughly an order of magnitude.

The interest in thermoelectric properties of ternary transition metal pnictides such as $Yb_{14}MnSb_{11}$ [30] and other ternary pnictides [30–34] with related structures such as the AZn_2Pn_2 (A=alkaline-earth metal) phases prompted us to carry out measurements of Seebeck coefficient and thermal conductivity of $BaCu_2As_2$. The electrical resistivity values (Fig. 6(a)) for this pellet is ≈9 times higher than those measured on a single crystal

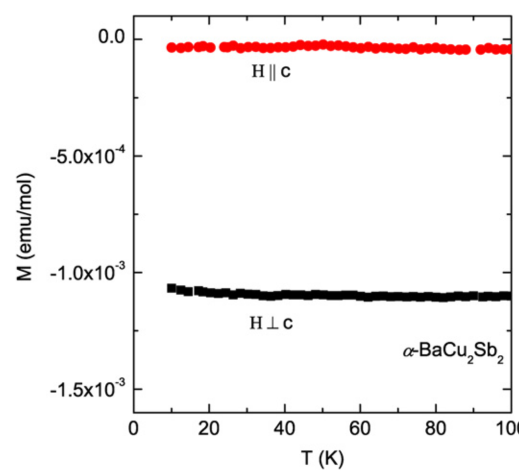

Fig. 4. Temperature dependence of magnetization along and perpendicular to the *c*-axis below 100 K for α-$BaCu_2Sb_2$.

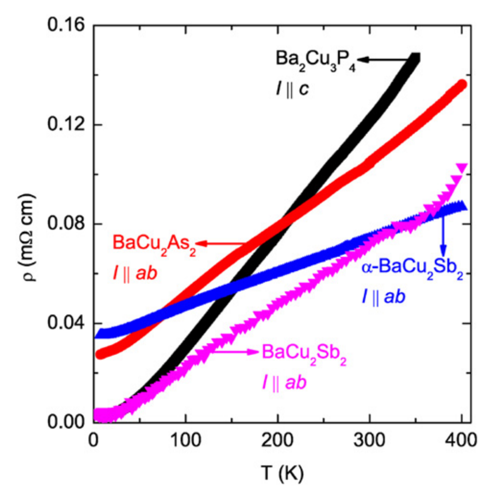

Fig. 5. Temperature dependence of electrical resistivities of $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ (Pn=As, Sb) single crystals. For $Ba_2Cu_3P_4$, $BaCu_2As_2$ and α-$BaCu_2Sb_2$, flux grown single crystals were measured. For $BaCu_2Sb_2$, melt grown single crystals were used. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

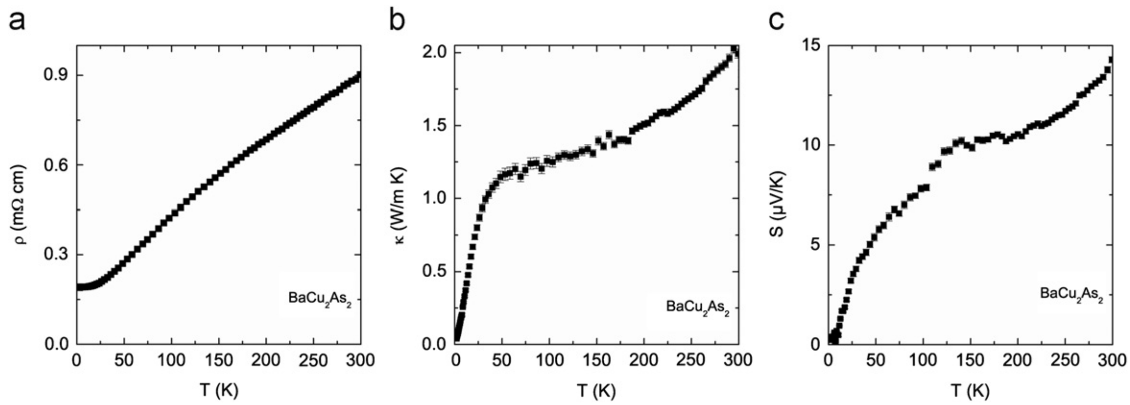

Fig. 6. Temperature dependence of (a) electrical resistivity, (b) thermal conductivity, and (c) Seebeck coefficient for $BaCu_2As_2$ polycrystalline sample.

(Fig. 5), which gives an estimate of anisotropy. The room temperature resistivity is ≈ 0.91 mΩ cm, while the RRR is ≈ 4.7. The thermal conductivity (Fig. 6(b)) has nearly even electronic and lattice contributions in 100–200 K range, with the calculated electronic contribution reaching ≈ 0.71 W/(m K) at 200 K. This value is calculated using the Wiedemann–Franz law [35], which states that the ratio of the electronic contribution to the thermal conductivity (κ_e) and the electrical conductivity (σ) of a metal is proportional to the temperature (T), i.e., $\kappa_e/\sigma = LT$ (where L is the Lorenz number, 2.44×10^{-8} W Ω/K^2). The low temperature peak in thermal conductivity curve, which is typical for crystalline materials due to umklapp phonon–phonon interactions [35], is suppressed, likely due to the high electrical conductivity values and presence of defects in this material [36–37]. The curvature above 200 K in the temperature dependence of the thermal conductivity is due to the radiative heat losses, which is an artifact of the measurement method used (see experimental section). The numerical values of thermal conductivity for $BaCu_2As_2$ are quite low (≈ 1.5 W/(m K) at 200 K), for comparison, the room temperature values of ≈ 0.8 W/(m K) for $Yb_{14}MnSb_{11}$ [30] and 1.34–4.25 W/(m K) for the solid solutions $Ca_xYb_{1-x}Zn_2Sb_2$ [31] have been reported for the few promising thermoelectric materials. The thermal conductivity at room temperature in $BaCu_2As_2$ could possibly be further lowered by introducing defects, alloying, and preparing solid solutions through gradual substitution of As with Sb. The Seebeck coefficients are also quite low for this material, with room temperature value of ≈ 15 μV/K. Consequently, in the measured interval, the highest figure-of-merit (zT) value (calculated using the formula $zT = (S^2T)/(\rho\kappa)$, where S is the Seebeck coefficient, T is the absolute temperature, ρ is the electrical resistivity, and κ is the total thermal conductivity) is only ≈ 0.03.

Due to a larger cell volume and more complex structure with varying layers, perhaps $BaCu_2Sb_2$ (both α- and β-forms) would have been more interesting choice for study of thermoelectric properties. But, when a cold-pressed pellet of the α-modification is heated, a secondary β-form, and possibly other polymorphs appear in the sample. So, the β-form itself could not be obtained as a pure phase (Fig. 3), which prevented thermoelectric studies on single phase α- and β-forms of $BaCu_2Sb_2$. Notwithstanding this fact, all studied modifications of $BaCu_2Sb_2$ are metals, and therefore, low Seebeck coefficients are expected for them as well.

4. Conclusions

Three ternary copper pnictides of barium, $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ (Pn = As, Sb) have been synthesized, and their magnetic, electrical, and thermal transport properties studied. The synthetic efforts using Pb flux resulted in a discovery of a new polymorph of $BaCu_2Sb_2$, namely the α-modification, which adopts $CaBe_2Ge_2$-type structure. The structural studies employing single crystal and powder X-ray diffraction methods suggest, although related, differing crystal structures. In the case of $BaCu_2Sb_2$, hints of other polymorphs have been also observed.

The studied compounds behave as metals, in an agreement with the earlier theoretical work on $SrCu_2As_2$ and $BaCu_2As_2$. The electrical resistivity values decrease from 0.08 to 0.15 mΩ cm at room temperature down to less than 0.04 mΩ cm at 2 K. No apparent trend based on the sizes and electronegativities of pnictogen atoms has been observed for the investigated phases, most likely due to the different crystal, and hence, electronic structures. The theoretical work on $BaCu_2As_2$ [11] also predicts that these compounds are far from magnetism, which was confirmed by our magnetization measurements on single- and poly-crystalline samples. The thermal conductivity of a cold-pressed pellet of $BaCu_2As_2$ was found to be quite low (≈ 1.5 W/(m K) at 200 K). However, the room temperature Seebeck coefficient value of ≈ 15 μV/K is also quite low, and therefore, based on the results of transport studies, $BaCu_2As_2$ is not expected to have high thermoelectric efficiency.

Further modifications on $BaCu_2As_2$, and especially measurements on clean $BaCu_2Sb_2$ samples may provide even lower thermal conductivity values. However, there is no indication that the studied compounds could develop superconducting behavior, because as suggested by the results of the electronic structure calculations [11], the compounds behave very differently compared to the analogs containing Fe, Co and Ni both in terms of magnetism and electrical transport properties. Additionally, the presence of strong interlayer bonds in $Ba_2Cu_3P_4$ and $BaCu_2Pn_2$ is a significant disadvantage for consideration of these phases for further investigations, as it has been shown that such bonds in $CaFe_2P_2$ prevent necessary nesting conditions for superconductivity [13]. The Fermi surface topology of $CaFe_2P_2$ is very different compared to that of superconducting pnictides, in turn, it resembles the Fermi surface of the non-superconducting collapsed tetragonal $CaFe_2As_2$, which is obtained by applying hydrostatic pressure [13]. Although the electronic structure of $CaFe_2As_2$ under zero pressure is similar to that of superconducting pnictides, in the collapsed tetragonal $CaFe_2As_2$, the c-axis is decreased through formation of the As–As bonds, which leads to dramatic changes in the Fermi surface of the phase.

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